Unearthing Drivers Behind Mortality: Exploring Global Patterns in Causes of Death (1950–2019)

1st Ronodeep Bhowmik
Dept of CSE
BRAC University
Dhaka, Bangladesh
ronodeep.bhowmik@g.bracu.ac.bd

2nd Anwesha Roy
Dept of CSE
BRAC University
Dhaka, Bangladesh
anwesha.roy@g.bracu.ac.bd

2nd Fatema Khanom
Dept of CSE
BRAC University
Dhaka, Bangladesh
fatema.khanom@g.bracu.ac.bd

***Abstract*—This paper constructs a machine learning model to predict national epidemiologic shifts with historical mortality data. Based on a dataset of causes of death word all over the world (1990-2019), with a total of 204 countries, we fabricated eight proxy characteristics of socio-economic, demographic, and health-related variables. A Random Forest predictor was defined to identify the dominant disease profile, such as Chronic-Dominant, Infectious-Dominant, or Transitional, in individual countries in 2010-2019 using the historical trends of the same 1990-2009. In a set of silent data, the optimal model reached a balanced memory of 67.8 and especially a high performance on detecting countries with an infectious predominance. Although there are still some limitations in conveying the differences between chronic and transitional profiles, the findings demonstrate where mortality data can be used as an indicator of future health planning priorities by the world. This work illustrates the use of machine learning to transform past health records into informative outputs that can be used by the policy makers and other researchers.**

***Index Terms*—Mortality trends, epidemiological transition, machine learning, Random Forest, global health**

I. INTRODUCTION

Introduction Mortality trends are regarded as one of the most significant indicators of global health and human development. Understanding the mortality figures and exploring the correlating reasons provides valuable information on the socio-economic and socio-medical aspects of any society. “Unearthing Drivers Behind Mortality: Exploring Global Patterns in Causes of Death (1990 – 2019)” is focused on a large database covering 204 countries and territories over a period of 30 years. During the past few decades, world health has changed remarkably. As the WHO and GBD Study describe, many low and middle-income countries are, with increases in aging and urbanization, adopting new styles and experiencing a shift from a predominance of infectious to non-communicable, lifestyle-related diseases. For instance, the Global Burden of Disease (GBD 2019) study shows that the number of deaths caused by cardiovascular diseases (CVD) has appreciated from 12.1 million in 1990 to 18.6 million in 2019, and remains the single greatest global killer.As the WHO highlights, in 2022, approximately 32 % of deaths worldwide, in low- and middle-income countries, was due to cardiovascular diseases with more than three-quarters of such deaths being CVDs [2]. Although mortality from diseases such as HIV/AIDS and malaria—once among the primary causes—has decreased as a result of global health initiatives, expanded vaccination campaigns, and enhanced access to treatment, emerging threats like air pollution, substance use disorders, and increasing obesity rates pose new challenges for public health infrastructures. Within this context, our project exceeds descriptive statistics. By subjecting the dataset to machine learning techniques, it extracts the characteristic ”disease signature” of every country and analyzes how historical mortality profiles (1990–2009) can predict leading causes in a subsequent decade (2010–2019). The dataset aggregates deaths across 31 major causes, ranging across non-communicable diseases (e.g., cardiovascular diseases, cancers, dementia) to communicable diseases (e.g., malaria, tuberculosis, diarrheal diseases) and injury-related or external causes (e.g., drowning and interpersonal violence).By converting raw numbers this work, has contributed to organizing, in comparable country-year profiles, information. brings a new chance of finding out the deter- shifting. minants of global mortality. Such a predictive model not only. registers diminished changes in commanding health but so also easi- mates uncertainties and discloses foremost explanatory factors. behind these shifts. This kind of methodology can be used best. to nations with so-called dual burden of disease- concomitant presence or exposure to endemic infections. and increasing non-communicable health threats. The ultimate aim is to massage uncooked mortality data into decision-intelligible infullation. gence. The concentration of deaths was analyzed to give a change. in the immediate future, the work unveils practical evidence to. planners, policymakers and researchers. It supports resource sets the intervention priorities, drives intervention, and assists monitoring. movement in the right world directions like the United Nations. SDG 3 Goal (Good Health and Well-). Being), which demands the decrement of premature mortality out of. non-communicable diseases and cessation of preventable deaths. since there is contagious and maternal succession [3].

II. LITERATURE REVIEW

In the article [4] on hybrid time-series and machine-learning models to predict cardiovascular mortality, India’s IHME GBD data (1990-2021) were applied with five age groups to predict age-specific trends in heart disease mortality. It compared and contrasted the results of using RMSE and MAPE for the ARIMA and hybrid ARIMA-ML (Random Forest, SVM, XG-Boost, GARCH). The results have shown that hybrid frameworks, specifically, ARIMA+SVM and ARIMA+XGBoost, displayed a high degree of dependable execution, especially in accuracy (up to 15.6% of advancement). ARIMA+GARCH performed best in the mortality cases of 0-5; however, ARIMA+SVM can do better in the cases of 6-15 and 70 plus. Google has noted that despite the great predictive efficiency and continuity in computing this research, the study had some of the following problems: the dependency of the data on the research, overfitting, and the low interpretability of the results although the study had accomplished high predictive performance and efficiency of the computation, new areas of research ought to introduce external socioeconomic and environmental impacts in the research models so that validity of the models and relevancy of the policies can be good. The experiments, to which the authors apply the term in the article [5], entail placement of death toll foretelling of the various regions of the world irrespective of the task under forecast, education of the previously trained foundation models (TimesFM, CHRONOS) to similarly foretell the mortality prophesied besides death rates of the ARIMA, Exponential Smoothing, Random forest, LSTM, and Lee-Carter. It was helpful to use the SMAPE measure since it allowed comparing the 5-year projections with the 10-year forecast of 20-year projections, relying on the Human Mortality Database (50 countries and 111 age groups). They discovered that CHRONOS was a useful predictive modelling instrument in the short-run forecasting, and when configured, CHRONOS made colossal returns in long-run forecasting over its contemporary conventional requirements. The other likelihood is that the highest score in SMAPE was allocated to Random Forest. The inability to use pre-trained models is the greatest concern, followed by optimisation of hyperparameters and just one validation step. As a rule, we can say that the article demonstrates that the approaches of the zero-shot and machine-based can be utilised to improve the work of the mortality prediction and that the fine-tuning process should be undertaken, so that even the long-term and age-related predictions would be more accurate.

The DeepDeath method of estimating the underlying causal factors of death, using the open-ended American mortality data in the form of a massive one, is a deep learning model [6]. The 67 cause classes model was reached after the contributions of the projects had already been pre-programmed to remove the contributory and the infrequent causes. The similarities between DeepDeath as the classifier and two-layer LSTM to discover all the temporal relationships among sequential causes of death of the dead and the shallow models (uni-gram models, bi-gram models defined by Random Forest) were compared. These results indicated that DeepDeath by far outperforms any n-gram model and that the benefit of lapsing patterns by simply learning them instead of hand-coding their properties is feasible. The authors have also followed an additional step to ensure that features learned become more observable and represented them as t-SNE, which revealed the clusters of major exhaustive diseases. Although it was stated that deeply learning models cannot be easily deciphered they are not conducive to other methods of analysis and that future research of the generative models must be used to help complete the death certificates it is stated the paper that deep learning is a factual working tool that can come in handy any given opportunity mortality data that is rather complex are those that need to be filled into the death certificates. A variety of machine learning algorithms were applied to 651 Bangladesh residents in this paper [7] in predicting cardiovascular disease (CVD) using rich socio-demographic, lifestyle and clinical characteristics. The joint logistic regression, decision trees, and other ensemble activities of performance could not compete with the overall performance of the random Forest, which was 98.04, 100, and 0.989, respectively. A comparatively strong risk impact of cholesterol was demonstrated by SHAP, which stated that cholesterol, hypertension, irregular heart rhythm, sleep apnea, family history, and signs of mental health are significant predictors. Though the results indicate that machine learning is capable of predicting CVD rather well, and that the application of Random Forest in this specific case would be suitable, the presented study was not longitudinal and utilised a rather restricted set of subjects, limiting the appropriateness of the findings obtained to be applied to social health planning in the future. The experiment [8] evaluates several machine learning methods to forecast infant mortality in Bangladesh by using socio-demographic, maternal, and environmental predictors based on the BDHS 20172018 survey data (26145 records). The conclusion was made in the SVM and the Logistic Regression compared with Decision Tree, Random Forest (RF) and the SVM based on accuracy, sensitivity, specificity, F1-score, ROC, with Boruta test and chi-square test being used as criteria of feature selection. It was also noted that RF was generally quite effective in predicting and generalising across folds of cross-validation because it achieved the highest predictive accuracy and generalisation, but the Logistic Regression did not even do this with high overall accuracy. One point of fact is that educating parents, wealth index, maternal BMI, birth order and region were important predictors. As Expert Onion (2013) admits in the article, the advantage of RF in handling interaction between complex features is strong, but its black box nature makes it difficult to understand the processes, which a future study should investigate to offer a predictive potential, since one can understand the nature of the underlying models to respond to health recommendations.

III. Dataset Description

The dataset we used in our project was collected from Kaggle named “Cause of Deaths around the World (Historical Data)”. This data is a complete global panel of cause-of-death data ranging between the years 1990 and 2019 across 204 countries and territories. The total recorded deaths in all causes is 1,468,134,716. Organized in a tidy rectangular format, a single row corresponds to a particular territory or country in a given year, while a column registers numbers deceased due to leading categories of diseases. In total, the dataset comprises a number of observations (total) 6,120 and total number 34 is suitable both to exploration and predictive work.There is 0 missing value in all the columns. The schema is arranged in three metadata fields: Coun- try/Territory, Code, and Year, 31 cause-specific variables. These include pre-eminent non-communicative dis- eases like Cardiovascular Diseases, Chronic asthmatic. Health-related problems Tuberculosis, HIV/AIDS, Malaria and Diarrheal Diseases etc. The numbers reported in these columns are raw non-negative integers of death. By data quality, the data set has a small number of salient strengths. It has no unfavorable or unreported measurements in the mortality characteristics and therefore, ensures integrity and preciseness so that it can be modeled at the statistical level. Its clean-panel the structure facilitates the easy aggregation (make global), to such an extent that formation permits it to calculate time trend research, sums or geographic distributions (See long-run decreases or increases of the selected mortality. Prepare country-level profiles, or causes). One can even create data-mining predictive models where years of the past are the predictor. Due to reasons unknown, variables and following years work as measures of response to the clear time period. This information is of infrastructural significance to the sphere of world health policy and research of health conditions. It supports the study of the epidemiological transition and the work. The remaining historical distinctive subregional disparities in causes of death. With thick coverage and far-reaching reviews of quality, the data is dense. Something reliable to draw conclusions out of based on data that trends of global deaths.

The experiment evaluates several machine learning methods to forecast infant mortality in Bangladesh by using socio-demographic, maternal, and environmental predictors based on the BDHS 2017-2018 survey data (26145 records). The conclusion was made in the SVM and the Logistic Regression compared with Decision Tree, Random Forest (RF) and the SVM based on accuracy, sensitivity, specificity, F1-score, ROC, with Boruta test and chi-square test being used as criteria of feature selection. It was also noted that RF was generally quite effective in predicting and generalising across folds of cross-validation because it achieved the highest predictive accuracy and generalisation, but the Logistic Regression did not even do this with high overall accuracy. One point of fact is that educating parents, wealth index, maternal BMI, birth order and region were important predictors. As Expert Onion (2013) admits in the article, the advantage of RF in handling interaction between complex features is strong, but its black box nature makes it difficult to understand the processes, which a future study should investigate to offer a predictive potential, since one can understand the nature of the underlying models to respond to health recommendations.

IV. Data Preprocessing

The raw dataset underwent validation checks to verify relevance for analysis. The columns were subdivided into an identifier set Country/Territory,Code, Year) and 31 cause-ofdeath attributes, and checks verbalized no blank and no negative numbers in either of the death columns. The validation step let the data set available to both descriptive and predictive analysis yet, with not an augmentation of cleaning or imputation. To facilitate temporal prediction, years by year two divisions were made on the dataset. One is from 1990-2009 to compose the historical features and the other is 2010-2019 in deriving future target labels.

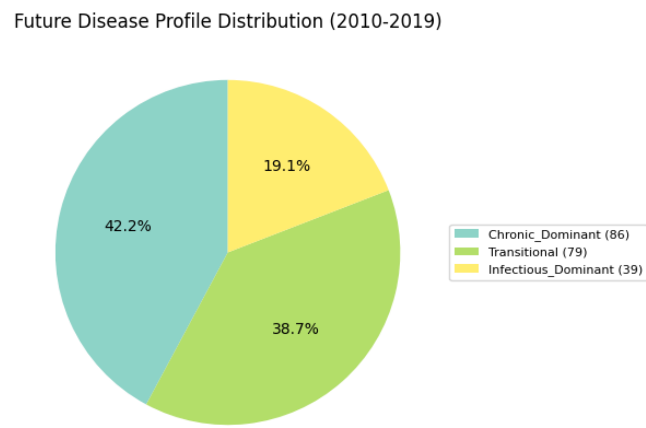

Fig. 1. Future Disease Profile Distribution (2010–2019).

Future labels were defined by averaging deaths per cause over the period 2010-2019 for each country and identifying the top three causes of death. Based on the composition of these top causes, countries were classified into one of three epidemiological profiles:

- **Chronic_Dominant_Future**: All top-3 causes are chronic diseases
- **Infectious_Dominant_Future**: At least two infectious or maternal causes, or a combination of infectious and maternal causes
- **Transitional_Future**: A mixed pattern of chronic and infectious causes

Beyond the raw counts, **semantically rich features** were engineered from historical cause vectors to capture higher-level structural patterns of mortality. These included:

- **Historical_Development_Proxy**: Ratio of chronic to infectious deaths
- **Historical_Healthcare_Quality_Proxy**: Share of preventable deaths
- **Historical_Age_Structure_Proxy**: Ratio of elderly- versus young-associated causes
- **Historical_Violence_Proxy**: Proportion of deaths due to violence, conflict, or self-harm
- **Historical_Environmental_Proxy**: Burden of tropical or environmental causes

- **Historical_Deaths_Per_Million**: Scale proxy based on total deaths
- **Historical_Baseline_Year**: A constant reference year (1999.5)
- **Historical_Mortality_Diversity**: Entropy of the cause distribution, reflecting the spread of mortality across causes

The last step in processing was the construction of a clean, country-level table containing the three-class future profile as the target and eight engineered historical features as the inputs. This table was ready for stratified train, validation, and test splits, supporting robust model training while preserving class balance and ensuring interpretability of the features in predicting future epidemiological patterns.

V. Modeling Approach

This study casts the problem of forecasting future population health trends as a multi-class supervised classification task. The objective is to classify a country's dominant disease profile during the target period (2010–2019) using only historical cause-of-death data (1990–2009), thereby capturing stages of epidemiological transition through data-driven inference.

A. Problem Formulation

- **Input Features (X):** Historical mortality patterns from 1990–2009, aggregated into eight engineered features that serve as proxies for socio-economic, demographic, and public health conditions. These features move beyond raw death counts to capture higher-level trends.
- **Target Variable (y):** A categorical variable representing the future disease profile of a country, derived from the top three causes of death during 2010–2019. The three classes are:
  1) **Chronic-Dominant Future:** Top causes are non-communicable diseases (e.g., cardiovascular diseases, neoplasms)
  2) **Infectious-Dominant Future:** Dominated by infectious, maternal, or neonatal causes (e.g., HIV/AIDS, malaria, neonatal disorders)
  3) **Transitional Future:** A mixed profile of chronic and infectious diseases

B. Feature Engineering

To prevent overfitting and enhance interpretability, eight predictive features were constructed from the historical data:

- **Historical_Development_Proxy:** Ratio of chronic to infectious disease deaths
- **Historical_Healthcare_Quality_Proxy:** Percentage of deaths from preventable causes (e.g., maternal and nutritional deficiencies)
- **Historical_Age_Structure_Proxy:** Ratio of elderly-related deaths to youth-related deaths
- **Historical_Violence_Proxy:** Proportion of deaths from violence, self-harm, or conflict
- **Historical_Environmental_Proxy:** Percentage of deaths from tropical diseases (e.g., malaria, diarrheal diseases)
- **Historical_Deaths_Per_Million:** Scaled total deaths, serving as a proxy for population size and overall mortality
- **Historical_Baseline_Year:** Fixed temporal reference (midpoint of training period)
- **Historical_Mortality_Diversity:** Entropy-based measure of cause-of-death diversity

C. Algorithm Selection

A **Random Forest classifier** was selected for this task due to its:

- Capacity to model **non-linear relationships** between features and classes
- Native support for **multi-class classification**
- Robustness against overfitting through ensemble learning
- Provision of **feature importance scores**, enabling interpretation of key predictors
- Effectiveness with tabular data containing engineered features

This approach combines epidemiological theory with machine learning rigor, using engineered proxies to transform complex historical data into actionable insights for predicting future health burdens.

VI. Model Training and Hyperparameter Tuning

A. Data Splitting Strategy

To ensure robust evaluation and prevent overfitting, the dataset of 204 countries was partitioned into three subsets using stratified sampling to maintain the proportional distribution of target classes across all splits:

- **Training Set (60%, 122 countries):** Used to train the Random Forest model
- **Validation Set (20%, 41 countries):** Used for hyperparameter tuning and model selection during grid search
- **Test Set (20%, 41 countries):** Held out for final evaluation, providing an unbiased estimate of model performance on unseen data

B. Hyperparameter Tuning

A comprehensive grid search was conducted to optimize the Random Forest classifier:

- **Method:** GridSearchCV with 5-fold cross-validation
- **Scoring Metric:** balanced_accuracy (selected to account for class imbalance)

- **Parameter Grid:**
  - *n_estimators:* [100, 200, 300] (number of trees in the forest)
  - *max_depth:* [10, 15, 20, None] (maximum depth of trees)
  - *min_samples_split:* [2, 5, 10] (minimum samples required to split a node)
  - *min_samples_leaf:* [1, 2, 4] (minimum samples required at a leaf node)
  - *max_features:* ['sqrt', 'log2', None] (number of features considered for splitting)
- **Search Process:** The grid search evaluated 324 parameter combinations using cross-validation on the training set. The best-performing model was selected based on the highest mean cross-validation balanced accuracy score.

C. Final Model Configuration

The optimal hyperparameters identified were:

- *n_estimators:* 300
- *max_depth:* 10
- *min_samples_split:* 10
- *min_samples_leaf:* 4
- *max_features:* None

The tuned model achieved a cross-validation balanced accuracy score of **0.8271** during training, indicating strong performance before final evaluation on the test set. This configuration was used for all subsequent validation and testing phases to ensure consistent and reproducible results.

VII. Evaluation Metrics

A comprehensive suite of evaluation metrics was employed to rigorously assess the performance of the optimized Random Forest classifier. This multi-faceted approach was necessary to provide a holistic view of the model's capabilities, particularly due to the imbalanced nature of the target classes. Performance was evaluated on a strictly held-out test set of 41 countries that was never utilized during training or hyperparameter tuning, ensuring an unbiased assessment of generalizability.

A. Primary Performance Metrics

1) Accuracy: Accuracy measures the proportion of total predictions that are correct, calculated as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

The model achieved a test **accuracy of 63.4%**, correctly predicting the future disease profile for 26 out of 41 countries in the test set.

2) Balanced Accuracy: To address class imbalance (Chronic-Dominant: 42.2%, Transitional: 38.7%, Infectious-Dominant: 19.1%), Balanced Accuracy was used as the primary metric. It represents the macro-average of recall scores per class:

$$\text{Balanced Accuracy} = \frac{1}{C}\sum_{i=1}^{C}\text{Recall}_i$$

where C is the number of classes. The model's **balanced accuracy of 67.8%** indicates more consistent performance across all classes than raw accuracy suggests.

B. Secondary Metrics

Weighted averages were computed for precision and F1-score to incorporate class imbalance by weighting each class's score by its support.

- **Precision (Weighted: 0.6516):** Measures the proportion of correct positive predictions among all positive predictions
- **F1-Score (Weighted: 0.6315):** Harmonic mean of precision and recall, providing balance between false positives and false negatives

C. Confusion Matrix Analysis

The confusion matrix provides detailed insight into prediction patterns:

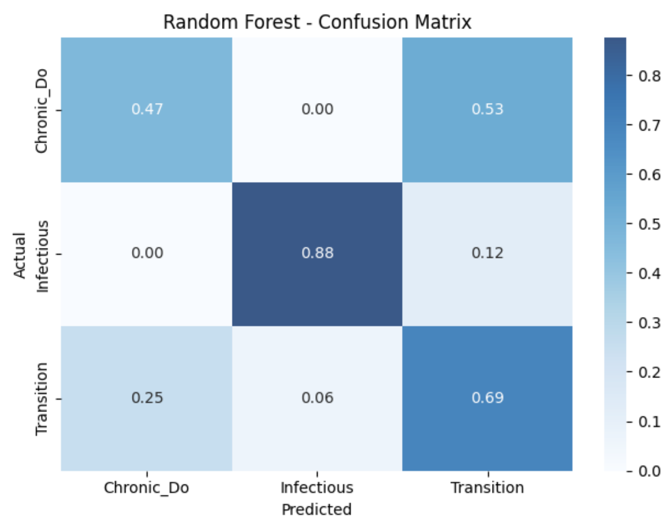

Fig. 2. Confusion Matrix.

Key observations from the confusion matrix:

- **Infectious-Dominant Class:** Excellent performance with recall of 0.88 (7/8 correct), crucial for identifying high-burden countries
- **Chronic-Dominant Class:** 8 correctly identified, but 9 confused with Transitional, suggesting similar historical patterns
- **Transitional Class:** Most challenging class with 11 correct, but 4 confused with Chronic-Dominant and 2 with Infectious-Dominant

The evaluation metrics collectively indicate that the model has **moderate predictive power** with a test balanced accuracy of 67.8%. The strong performance on the Infectious-Dominant class (recall: 0.88) demonstrates particular utility for identifying vulnerable populations. The primary challenge lies in distinguishing between Chronic-Dominant and Transitional futures, an expected difficulty given the continuous nature of epidemiological transition. The model provides a valuable, data-driven tool for forecasting population health trajectories with performance well-substantiated for a real-world, imbalanced classification problem.

VIII. Results and Performance

The optimized Random Forest model was rigorously evaluated on both a validation set, used for model selection, and a final held-out test set, which provides an unbiased estimate of its real-world performance. The results demonstrate the model's capability to predict future epidemiological transitions based on historical data.

A. Validation Set Performance

The validation set, comprising 41 countries, was used to fine-tune the model and select the best hyperparameters. Performance on this set indicates how well the model generalized to unseen data during the development phase.

TABLE I
Model Performance on Validation Set (41 Countries)

Metric	Score	Interpretation
Validation Accuracy	78.6%	Correctly predicted future profile for ∼32 of 41 validation countries.
Validation Balanced Accuracy	81.6%	Maintained strong performance across all three classes despite imbalance.

The high balanced accuracy on the validation set confirmed that the chosen hyperparameters were effective and that the model was not overfitting to the training data.

B. Test Set Performance

The final evaluation was conducted on a completely unseen test set of 41 countries. The results on this set are the definitive measure of the model's predictive power.

TABLE II
Final Model Performance on Held-Out Test Set

Metric	Score
Test Accuracy	63.4%
Test Balanced Accuracy	67.8%
Test Precision (Weighted)	0.6516
Test F1-Score (Weighted)	0.6315

TABLE III
Detailed Classification Report by Class on Test Set

Class	Precision	Recall	F1-Score	Support
Chronic-Dominant Future	0.67	0.47	0.55	17
Infectious-Dominant Future	0.78	0.88	0.82	8
Transitional Future	0.52	0.65	0.58	16
Macro Avg	0.66	0.66	0.65	41
Weighted Avg	0.65	0.63	0.63	41

C. Analysis of Results

1) Overall Performance: The model achieved a **test accuracy of 63.4%** and a more informative **balanced accuracy of 67.8%**. This represents reasonable predictive capability for a complex, real-world problem involving forecasting future trends from historical patterns.

2) Class-Wise Performance: The model excelled at identifying countries with an **Infectious-Dominant Future**, achieving high precision (0.78) and recall (0.88). This is a critical strength, as accurately pinpointing nations with a high burden of preventable diseases is highly valuable for public health planning.

3) Primary Challenge: The key point of error was mixed-upness between the Chronic-Dominant and Transitional classes, are supported by the decreased precision and recall of these categories. A natural and rational difficulty this is, as historical mortality of these two groups is properly more like one another than unlike pattern of Infectious-Dominant group. The model is effective in terms of historical mortality proxies to predict general epidemiological changes with moderate to good accuracy. It is particularly robust at detecting nations which still remain in infectious-dominant fresh-made upholsters stage, and the difference between transitional and fully transitional states is a more finer, though not yet informative, prediction challenge.

The model successfully leverages historical mortality proxies to forecast broad epidemiological transitions with moderate to good accuracy. It is particularly robust at identifying countries that remain in the infectious-dominant stage, while the distinction between transitional and fully transitioned states presents a more nuanced, though still informative, prediction challenge.

IX. Conclusion

This study developed a machine learning framework to predict national epidemiological transitions based on historical mortality data. By analyzing cause-of-death patterns from 1990–2009, we successfully forecasted countries' dominant disease profiles for 2010–2019, classifying them into three distinct transition stages: Chronic-Dominant, Infectious-Dominant, and Transitional futures.

Through strategic feature engineering, we transformed raw mortality data into eight interpretable proxy variables representing socio-economic and demographic trends. An optimized Random Forest classifier, tuned to address class imbalance, effectively modeled the relationship between these historical proxies and future outcomes.

The model achieved a test balanced accuracy of 67.8%, with particularly strong performance in identifying Infectious-Dominant countries (recall: 0.88). This demonstrates the significant predictive value of historical mortality patterns as indicators of future health burdens. While some confusion occurred between Chronic-Dominant and Transitional categories—reflecting the continuous nature of epidemiological transition—the overall results provide policymakers with a valuable tool for anticipating disease burden shifts and prioritizing resource allocation.

The findings affirm that machine learning can extract meaningful, predictive signals from complex public health data, translating historical patterns into actionable foresight. Future research could enhance predictions by incorporating additional economic and healthcare access indicators, potentially offering deeper insights into the drivers of global mortality transitions.

X. Limitation

This study has a few limitations that affect how the results should be understood. The labels of the future were created by linking the top three causes of death, which can lead to oversimplifying the mortality complex and make the results overly sensitive to minor shifts in ranking. Additionally, the large aggregation windows of 20 years of historical data with 10 years of future data introduce bias by smoothing differences in data, thus making the data uniform. Furthermore, the only mortality data present was collected without the external factors of demography, economy, and health care, which limits the insight and strength of the predictive model.

In the case of the model, there is a lack of scope. The Random Forest model, which has a good range of accuracy as well as feature importance, does not possess causal interpretability, counterfactual reasoning, and probability calibration. There is also the issue of class imbalance, in which the lower strata are more difficult to predict. And the temporal setup only predicts the period that has been observed, not the actual future, which is a major limitation. Finally, the ease with which the data can be compared from country to country can lead to bias, and the lack of uncertainty quantification, such as confidence intervals or robustness checks, does not allow for the claims of the data to be very stable.

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